Overview of Actor critic algorithm

Subject: Actor critic algorithm

1.

Variants Asynchronous Advantage Actor-Critic (A3C): Parallel and asynchronous version of A2C. Soft Actor-Critic (SAC): Incorporates entropy maximization for improved exploration. Deep Deterministic Policy Gradient (DDPG): Specialized for continuous action spaces.

2.

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3.

Critic In the unbiased estimators given above, certain functions such as $V^{\pi _{\theta }},Q^{\pi _{\theta }},A^{\pi _{\theta }}$ appear. These are approximated by the critic. Since these functions all depend on the actor, the critic must learn alongside the actor. The critic is learned by value-based RL algorithms. For example, if the critic is estimating the state-value function $V^{\pi _{\theta }}(s)$, then it can be learned by any value function approximation method. Let the critic be a function approximator $V_{\phi }(s)$ with parameters ϕ . The simplest example is TD(1) learning, which trains the critic to minimize the TD(1) error: $\delta _{i}=R_{i}+\gamma V_{\phi }(S_{i+1})-V_{\phi }(S_{i})$ The critic parameters are updated by gradient descent on the squared TD error: $\phi \leftarrow \phi -\alpha \nabla _{\phi }(\delta _{i})^{2}=\phi +\alpha \delta _{i}\nabla _{\phi }V_{\phi }(S_{i})$ where α is the learning rate. Note that the gradient is taken with respect to the ϕ in $V_{\phi }(S_{i})$ only, since the ϕ in $\gamma V_{\phi }(S_{i+1})$ constitutes a moving target, and the gradient is not taken with respect to that. This is a common source of error in implementations that use automatic differentiation, and requires "stopping the gradient" at that point. Similarly, if the critic is estimating the action-value function $Q^{\pi _{\theta }}$, then it can be learned by Q-learning or SARSA. In SARSA, the critic maintains an estimate of the Q-function, parameterized by ϕ , denoted as $Q_{\phi }(s,a)$. The temporal difference error is then calculated as $\delta _{i}=R_{i}+$

γ Q θ (S i + 1 , A i + 1) − Q θ (S i , A i) {\displaystyle \delta _{i}=R_{i}+\gamma Q_{\theta }(S_{i+1},A_{i+1})-Q_{\theta }(S_{i},A_{i})} . The critic is then updated by θ ← θ + α δ i ∇ θ Q θ (S i , A i) {\displaystyle \theta \leftarrow \theta +\alpha \delta _{i}\nabla _{\theta }Q_{\theta }(S_{i},A_{i})} The advantage critic can be trained by training both a Q-function Q φ (s , a) {\displaystyle Q_{\phi }(s,a)} and a state-value function V φ (s) {\displaystyle V_{\phi }(s)} , then let A φ (s , a) = Q φ (s , a) − V φ (s) {\displaystyle A_{\phi }(s,a)=Q_{\phi }(s,a)-V_{\phi }(s)} . Although, it is more common to train just a state-value function V φ (s) {\displaystyle V_{\phi }(s)} , then estimate the advantage by A φ (S i , A i) ≈ ∑ j ∈ 0 : n − 1 γ j R i + j + γ n V φ (S i + n) − V φ (S i) {\displaystyle A_{\phi }(S_{i},A_{i})\approx \sum _{j\in 0:n-1}\gamma ^{j}R_{i+j}+\gamma ^{n}V_{\phi }(S_{i+n})-V_{\phi }(S_{i})} Here, n {\displaystyle n} is a positive integer. The higher n {\displaystyle n} is, the more lower is the bias in the advantage estimation, but at the price of higher variance. The Generalized Advantage Estimation (GAE) introduces a hyperparameter λ {\displaystyle \lambda } that smoothly interpolates between Monte Carlo returns (λ = 1 {\displaystyle \lambda =1} , high variance, no bias) and 1-step TD learning (λ = 0 {\displaystyle \lambda =0} , low variance, high bias). This hyperparameter can be adjusted to pick the optimal bias-variance trade-off in advantage estimation. It uses an exponentially decaying average of n-step returns with λ {\displaystyle \lambda } being the decay strength.

4.

γ j ∑ n = 1 ∞ λ n − 1 1 − λ ⋅ (∑ k = 0 n − 1 γ k R j + k + γ n V π θ (S j + n) − V π θ (S j)) {\textstyle \gamma ^{j}\sum _{n=1}^{\infty }{\frac {\lambda ^{n-1}}{1-\lambda }}\cdot \left(\sum _{k=0}^{n-1}\gamma ^{k}R_{j+k}+\gamma ^{n}V^{\pi _{\theta }}(S_{j+n})-V^{\pi _{\theta }}(S_{j})\right)} : TD(λ) learning, also known as GAE (generalized advantage estimate). This is obtained by an exponentially decaying sum of the TD(n) learning terms.

5.

γ j (∑ k = 0 n − 1 γ k R j + k + γ n V π θ (S j + n) − V π θ (S j)) {\textstyle \gamma ^{j}\left(\sum _{k=0}^{n-1}\gamma ^{k}R_{j+k}+\gamma ^{n}V^{\pi _{\theta }}(S_{j+n})-V^{\pi _{\theta }}(S_{j})\right)} : TD(n) learning.

6.

Actor The actor uses a policy function π (a | s) {\displaystyle \pi (a|s)} , while the critic estimates either the value function V (s) {\displaystyle V(s)} , the action-value Q-function Q (s , a) , {\displaystyle Q(s,a),} the advantage function A (s , a) {\displaystyle A(s,a)} , or any combination thereof. The actor is a parameterized function π θ {\displaystyle \pi _{\theta }} , where θ {\displaystyle \theta } are the parameters of the actor. The actor takes as argument the state of the environment s {\displaystyle s} and produces a probability distribution π θ (⋅ | s) {\displaystyle \pi _{\theta }(\cdot |s)} . If the action space is discrete, then ∑ a π θ (a | s) = 1 {\displaystyle \sum _{a}\pi _{\theta }(a|s)=1} . If the action space is continuous, then ∫ a π θ (a | s) d a = 1 {\displaystyle \int _{a}\pi _{\theta }(a|s)da=1} . The goal of policy optimization is to improve the actor. That is, to find some θ {\displaystyle \theta } that maximizes the expected episodic reward J (θ) {\displaystyle J(\theta)} : J (θ) = E π θ [∑ t = 0 T γ t r t] {\displaystyle J(\theta)=\mathbb {E} _{\pi _{\theta }}\left[\sum _{t=0}^{T}\gamma ^{t}r_{t}\right]} where γ {\displaystyle \gamma } is the discount factor, r t {\displaystyle r_{t}} is the reward at step t {\displaystyle t} , and T {\displaystyle T} is the time-horizon (which can be infinite). The goal of policy gradient method is to optimize J (θ)

{\displaystyle J(\theta)} by gradient ascent on the policy gradient ∇ J (θ) {\displaystyle \nabla J(\theta)} . As detailed on the policy gradient method page, there are many unbiased estimators of the policy gradient: ∇ θ J (θ) = E π θ [∑ 0 ≤ j ≤ T ∇ θ ln ■ π θ (A j | S j) · Ψ j | S 0 = s 0] {\displaystyle \nabla _{\theta }J(\theta)=\mathbb {E} _{\pi _{\theta }}\left[\sum _{0\leq j\leq T}\nabla _{\theta }\ln \pi _{\theta }(A_{j}|S_{j})\cdot \Psi _{j}{\Big |}S_{0}=s_{0}\right]} where Ψ j {\textstyle \Psi _{j}} is a linear sum of the following: